WEEK 2 — COMPARATIVE EXPERIMENT ANALYSIS

1. Introduction

The primary objective of this phase was to systematically evaluate different neural network configurations for the task of loan default prediction. The experiments were designed to investigate the impact of architectural choices, activation functions, optimization algorithms, regularization techniques, and decision thresholds on model performance.

Given the highly imbalanced nature of the dataset, where non-default cases significantly outnumber default cases, particular emphasis was placed on evaluating metrics beyond accuracy, such as precision, recall, and F1 score. These metrics provide a more meaningful assessment of model effectiveness in identifying defaulters.

2. Experimental Design

All experiments were conducted under controlled conditions to ensure consistency and comparability. The following parameters were kept constant unless explicitly modified:

- Dataset and preprocessing pipeline
- Train-validation-test split
- Number of epochs
- Batch size (except in specific experiments)
- Evaluation metrics

Each experiment altered only one variable at a time. This controlled approach ensured that any observed performance changes could be attributed directly to the modified parameter.

3. Phase I: Baseline Experiments (E1–E5)

3.1 Overview

The initial set of experiments focused on evaluating different neural network configurations without addressing class imbalance. These experiments included variations in model depth, activation functions, optimizers, and regularization techniques.

3.2 Observations

Across all baseline models, the following patterns were observed:

- Accuracy consistently remained high, approximately between 0.91 and 0.92
- Recall values were extremely low, ranging from approximately 0.006 to 0.044

- F1 scores were correspondingly low, indicating poor balance between precision and recall

3.3 Detailed Analysis

Experiment E1 — Baseline Model

The baseline multilayer perceptron with two hidden layers served as the reference model. While it achieved high accuracy, it demonstrated an almost complete inability to identify default cases. The model predominantly predicted the majority class.

Experiment E2 — Deeper Architecture

Increasing the depth of the network introduced additional representational capacity. However, the improvement in recall was marginal. This indicates that increasing model complexity alone does not address the fundamental issue of class imbalance.

Experiment E3 — Activation Function Change (ReLU to LeakyReLU)

The introduction of LeakyReLU was intended to improve gradient flow and prevent dead neurons. While there was a slight improvement in precision, the impact on recall remained minimal.

Experiment E4 — Optimizer Change (Adam to SGD)

The use of stochastic gradient descent resulted in poorer convergence, particularly for the minority class. Although precision increased, recall dropped further, indicating that the model became even more conservative in predicting defaulters.

Experiment E5 — Dropout Regularization

Dropout was introduced to reduce overfitting. While it improved generalization slightly, it did not significantly improve recall. The model still failed to capture minority class patterns.

3.4 Key Insight from Phase I

The results from baseline experiments clearly demonstrate that:

- High accuracy does not imply effective model performance in imbalanced datasets
- Architectural and hyperparameter changes alone are insufficient to address class imbalance
- The model is heavily biased toward predicting the majority class

4. Phase II: Class Imbalance Handling (E6)

4.1 Methodology

To address class imbalance, class weights were introduced during model training. This technique assigns higher importance to the minority class, forcing the model to pay more attention to defaulters.

4.2 Observations

- Recall increased dramatically from approximately 0.01 to 0.87
- Precision decreased significantly to approximately 0.11
- Overall accuracy dropped to approximately 0.43

4.3 Interpretation

The application of class weighting fundamentally altered model behavior:

- The model began actively identifying defaulters
- However, it also produced a large number of false positives
- The decrease in accuracy reflects the model's shift away from majority class dominance

4.4 Key Insight from Phase II

Class imbalance handling has a far greater impact on model performance than architectural modifications. It transforms the model from being ineffective to functionally useful, albeit with new trade-offs.

5. Phase III: Threshold Optimization (E7–E9)

5.1 Motivation

The default classification threshold of 0.5 is not always optimal, particularly in imbalanced classification problems. Adjusting the threshold allows control over the trade-off between precision and recall.

5.2 Observations

Threshold	Precision	Recall	F1 Score
0.3	0.113	0.874	0.200
0.4	0.135	0.764	0.229
0.5	0.159	0.642	0.256
0.6	0.185	0.518	0.273

5.3 Interpretation

- Lower thresholds increase recall but significantly reduce precision
- Higher thresholds improve precision but reduce recall
- Threshold 0.6 provides the best balance, as indicated by the highest F1 score

5.4 Key Insight from Phase III

Threshold tuning is a critical step in optimizing model performance. It allows fine-grained control over the balance between false positives and false negatives.

6. Comparative Summary

The progression of experiments can be summarized as follows:

- Baseline models achieved high accuracy but failed to detect defaulters
- Class weighting significantly improved recall but introduced many false positives
- Threshold tuning refined the model to achieve a balanced performance

7. Final Model Selection

The final model configuration (Experiment E9) was selected based on:

- Highest F1 score
- Acceptable balance between precision and recall
- Practical applicability in real-world scenarios

8. Limitations

Despite improvements, the model has several limitations:

- Precision remains relatively low, indicating a high number of false positives
- Class overlap in feature space limits separability
- Performance is sensitive to threshold selection

9. Conclusion

This study demonstrates that:

- Model evaluation must go beyond accuracy in imbalanced datasets
- Addressing class imbalance is essential for meaningful performance
- Threshold optimization plays a key role in achieving practical usability

The final model represents a significant improvement over the baseline and is suitable for applications where identifying defaulters is critical.